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QUARTERLY EXAMINATION - 2025

CLASS:XII | BIOLOGY | Reg.No

Time : 3.00 Hours | BIO-BOTANY | MAX MARKS : 70 | MARKS : 35

SECTION - I

Answer all the Questions : 8x1=8

Choose the most correct answer from the given alternate answer and write with its symbols.

1. Identify the incorrect pair.
   a) Sporopollenin - exine of pollen grain
   b) Tapetum - nutritive tissue for developing microspores
   c) Nucellus - nutritive tissue for developing embryo
   d) Obturator - directs the pollen tube into micropyle
2. Pick out an example to pleiotropy.
   a) Sickle cell anaemia b) ABO blood group system in human beings
   c) Skin colour in human d) Flower colour in Mirabilis jalapa
3. If haploid number in a cell is 18. The double monosomic and trisomic number will be
   a) 34 and 35 b) 34 and 37 c) 37 and 35 d) 17 and 19
4. Match the following :

Column - A	Column - B
1. Exonuclease	a. Add or remove phosphate
2. Endonuclease	b. Binding the DNA fragments
3. Alkaline Phosphatase	c. Cut the DNA at terminus
4. Ligase	d. Cut the DNA at middle

	1	2	3	4
A)	a	b	c	d
B)	c	d	b	a
C)	a	c	b	d
D)	c	d	a	b

5. Which of the following statement is correct?
   a) Agar is not extracted from marine algae such as seaweeds.
   b) Callus undergoes differentiation and produces somatic embryoids.
   c) Surface sterlization of explants is done by using 0.1% mercuric bromide.
   d) pH of the culture medium is 5.0 to 6.0
6. Solidifying agent used in plant tissue culture is
   a) Nicotinic acid b) Cobaltous chloride c) EDTA d) Agar
7. Which one of the following substance is generally absent around the germ pore of exine?
   a) Pollen kit b) Tryphine c) Special protein d) Sporopollinin
8. Choose the wrong pair of the following pairs.
   a) Polygenic inheritance - H. Nilsson - Ehle
   b) Pleiotropy - A single gene affects multiple traits
   c) Epistasis - blending inheritance
   d) Dihybrid cross - Law of independent assortment

SECTION-II

Answer any Four of the following 4x2=8

9. A leaf of Bryophyllum produces new plants. How?
10. Differentiate - genotype and phenotype.
11. What is called linkage?
12. What are the materials used to grow micro organism like spirulina?
13. What do you mean embryoids?
14. What are called callus?

SECTION - III

Answer any Three of the following. Quetion number 19 is compulsory. 3x3=9

15. Differentiate - grafting and layering.
16. What is called reciprocal cross? Write its importance.
17. What is called Genetic Map? Mention any two uses.
18. Name the chemicals used in gene transfer.
19. What is known as cryopreservation?

SECTION - IV

Answer all the questions. 2x5=10

20. a) Enumerate the characteristic features of Entomophilous flowers. **(OR)**
    b) Describe dominant epistasis with an example.
21. a) Write the benefits and risk of genetically modified foods. **(OR)**
    b) Write the applications of tissue culture.

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BIO-ZOOLOGY

Marks : 35

PART-I

8x1=8

Note : I. Answer all the Questions

II. Choose most appropriate answer from the given alternatives and write the Option code and corresponding answer

1. Who was the first to discover parthenogenesis reproduction in animals?
   a) Abraham Trumpley b) George Robertson c) Charles Bonnet d) Joshua Lederberg
2. The foetal membrane that forms the basis of the umbilical cord is
   a) Allantois b) Amnion c) Chorion d) Yolk sac
3. Protection of Children from Sexual Offences Act
   a) POCSO b) RMNCH c) RCH d) PCPNDT
4. Patau's Syndrome is also referred to as
   a) 13 Trisomy b) 18 Trisomy c) 21 Trisomy d) 17 Trisomy
5. The total number of nitrogenous bases in the human genome is estimated to be used
   a) 3.5 million b) 35000 c) 35 million d) 3.1 billion
6. The phenomenon of "industrial melanism" demonstrates?
   a) Natural selection b) Induced mutation c) Reproductive isolation d) Geographic isolation
7. AIDS virus has
   a) Single-stranded RNA b) Double-stranded RNA c) Single-stranded DNA d) Double-stranded DNA
8. BCG vaccine
   a) Typhoid b) Tuberculosis c) Polio d) Smallpox

PART-II

Answer any four of the following question.Question No. 14 is compulsory. 4x2=8

9. What is ovulation? On which day of the menstrual cycle does it occur?
10. Differentiate between foeticide and infanticide
11. Mention the symptoms of Down syndrome
12. What are operons? How many operon groups are there in E. coli?
13. List out the major gases seems to be found in the primitive earth
14. Why do you think it is not possible to produce a vaccine against the "common cold"?

PART-III

Answer any three of the following question 19 is compulsory 3x3=9

15. Placenta is an endocrine tissue.Justify
16. State Any three goals of the Human Genome Projects
17. Differentiate between convergent and divergent evolution with one example
18. What is the name of the hormone secreted by the thymus gland? Write two functions of that hormone
19. A trait in the body of the grandfather is passed on to the grandson through the daughter Give a flowchart for this type of inheritance

PART-IV

Answer all the question and draw pictures wherever necessary 5x2=10

20. Define infertility and write causes of infertility **[OR]**
    Explain the various phases of the menstrual cycle
21. Darwin's finches and Australian marsupials are suitable examples of adaptive radiation Justify the statement **[OR]**
    Suggest some ways to prevent the indiscriminate use of drugs and alcohol

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